

CART-470 Living Learning Contract

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Group Kanban Link:

<https://app.fizzy.do/6286165/public/boards/m3jX3AdrS55LtbHSKpwuCNjQ>

Design Problem as described by the client:

“How can we create an interface that allows editing and playing of a simple pre-existing game (e.g. Pong). The editing should allow a sufficient breadth of expressive possibilities that users can come up with genuinely diverse outcomes (e.g. aesthetics, mechanics, physics, sound design, etc.)”

Project Focus

- Focus on absurd deconstruction, rather than strictly a pedagogical tool for coding
- Come up with ideation prototypes for a potential product rather than developing the finished product itself
- Create a project interface that encourages experimentation
- **Have the final product reflect the client’s (Pippin Barr’s) ideologies associated with his research.**

Knowledge Base/ Fields/ Domains/ Keywords

- User Interface and User experience
- Various forms of prototyping
- Interactivity with unconventionality
- Experimental, chaotic thinking
- Coding in JavaScript with P5

Learning Objectives (Individual)

Olivia

Weekly Journal Link:

<https://github.com/olivax28/CART-470-Journal-Olivia-Axiuk/blob/main/CART470-OliviaAxiuk-ProjectJournal.md>

Objectives:

- Create engaging designs based on the client’s objectives
- Push my own creative limits, focusing on the client’s ideas of “absurd deconstruction”
- Push my knowledge and on the various skills gained from my time in the CART program (JavaScript, p.5, UI design, etc.) in a practical, unifying way
- Create and apply new organizational techniques in order to ease group work (Fizzy/Kanban)

Learning Activities:

- Become familiar with Pippin Barr's past work, such as his "iterative games"
- Explore the draft excerpts from Pippin Barr's book, provided by him in the group Discord discussion
- Do research on the potential behind the programs/ software that will potentially be used, such as Figma, the P5 library and the Unity game engine

Nat

Weekly Journal Link: <https://github.com/natninac/CART470/blob/main/Journal.md>

Objectives:

- Refine my project management skills
- Get more UX/UI design experience

Learning Activities:

- Research cognitive theories of learning
- Research game design literature
- Organize and co-organize team meetings as needed (agenda, task assignment, etc)

Julia

Weekly Journal Link: https://juliax28.github.io/Cart-470-Journal--Julia-Axiuk/Journals_470

Objectives:

- Push my sense of design to create something that breaks regular convention
- Challenge/refresh my p5 coding skills
- Familiarize myself with project organisation via Fizzy
- Develop more detailed brainstorming practices

Learning Activities:

- Analyze/play Pippin Barr's previous games. This will allow me to understand what types of challenges are often implemented in his work in order to direct the project into a direction that follows his philosophies.
- Read the excerpts from Pippin Barr's book

Nolan

Weekly Journal Link: <https://github.com/nolanbg/CART470Nolan/tree/main/ProcessJournal>

Objectives:

- Expand my own knowledge of game mechanics, design and philosophy outside of coding
- Apply my own programming skills where and if applicable
- Use areas I feel more strong in (graphic design, audio, writing) to create a more complete prototype from a design perspective.
- Improve my experience in the prototyping process
- Stay on top of weekly tasks to be a complete part of the team

Hanif

Weekly Journal Link:

https://github.com/haniflearnscoding/CART470_ProcessJournal/blob/main/entries/week-02/REA_DME.md

Objectives:

- **O1.** Apply Figma's interactive components, variables and conditional logic to build a clickable prototype of the game editor where at least three settings visibly change the game.
- **O2.** Test the prototype with non-coder classmates to check whether someone with no coding background can change a game on their own.
- **O3.** Revise the prototype after testing and keep a log of what changed and why.

Learning activities:

1. **Read Bret Victor's *Learnable Programming* and *Up and Down the Ladder of Abstraction* (O1, O3).** *Outcome:* a short reflection naming the design principles I'll apply to the prototype and to how I iterate on it.
2. **Review Pippin Barr's existing games (PONGS, Snakisms, Variations) (O1, O2).** *Outcome:* a list of what a player can change in each game, and the three settings the prototype will let users edit.

Learning Objectives (Group):

- Synthesizing our different art styles and skills
- Thinking about game-making and mechanics beyond coding
- Staying on top of planned weekly objectives
- Creating habits that allow for all members to stay organized and up to date

Learning Activities (Group)

1. **Read and respond to the Bret Victor essays** (linked in the contract as course readings).
 - *Learnable Programming* argues that learners need to see what a program is doing as they change it, that unlabeled numbers are a barrier, and that people should start with something on screen and react to it. It maps closely onto your prototype: sliders should be labelled and show their effect right away.
 - *Up and Down the Ladder of Abstraction* describes starting with the simplest version, controlling one parameter, stepping up to see patterns, then stepping back down to a concrete example. Use it to structure how you iterate.

- Write a short reflection on each (a page or less) and list three design principles you'll apply to your prototype. *(O1, O3)*
- 2. **Figma skill-building.** Build one small practice prototype for each tool (interactive component, variable, conditional) before using it in the project, for example a slider that changes a shape's size. *(O1)*
- 3. **Analyze Pippin's existing games** (PONGS, Snakisms, the Variations pages) and list which choices a player can change, to decide what your editor needs to expose. *(O2)*

Final Outcome Expectation:

1. **Visual prototype(s):** An interactive visual prototype that will allow the client to experience the user experience and overall feeling of the software
2. **Practical Prototype(s):** The actual “game” portion, whether or not incorporated with the visual prototype, will be a simplified version of what is pitched.
3. **Research:** Our final product should not only be a prototype, but should also come with documentation surrounding the ideas behind specific decisions made along the duration of the project.

Group Milestones

1. **Oct 21st** - For the midterm meeting, we should all have agreed upon the final idea, and have come up with an interactive Figma (or similar program) Prototype representing it
2. **Nov 4th** - Create a solid plan to work towards the final product, working out which parts should be programmed, which parts should remain a design and which parts should be combined, assign tasks beginning production
3. **Dec 2nd** - This final milestone represents the finalized project, all coded and interactive features should work smoothly without bugs and represent both our vision and that of the client's

WEEKLY PLAN

Week	Plan	Description	Work Due
1 SEPT 9			
2 SEPT 16			

Week	Plan	Description	Work Due
3 SEPT 23	Client Meeting	Meet Pippin, figure out a direction and scope for the project.	Questions & interview prep
4 SEPT 30	Individual design sketches will be discussed to further explore ideas	Lecture: Why we prototype (defining the problem space)	- Living learning contract - Crazy ideating
5 OCT 7	Have some individual visual prototypes with some interactivity done in order to continue discussions	While discussing different directions, we can each use methods of our own choice to bring in examples to discuss	- More solid Ideating (1 concept each)
6 OCT 21	Midterm Meeting	Have a “This is what we are actually doing” concept, a focused goal (that can still change, of course) that the whole team agrees on.	- One concept that we all agree on, basic interactivity on the ‘interactive’ prototype
7 OCT 28	Interactivity focus	With a more focused concept, we can begin a more coding-focused phase of the project	- Have a simple, working P5 project
8 NOV 4	Final Production Begins	Continuous production on the final project over the next few weeks, must be adaptable to include potential bugs, glitches and other obstacles	Continuous progress
9 NOV 11			Continuous progress

Week	Plan	Description	Work Due
10 NOV 18			Continuous progress
11 NOV 25			Continuous progress
12 DEC 2	Final Presentation	Create a final prototype for the client, synthesizing interactivity and design into one “playable” project. Due to the nature of the client’s wishes, it does not have to be a complete application, but must accurately reflect the client’s ideologies.	Final Interactive Prototype